

Exp : 4 Backup Automation using Azure Blob Storage

Aim

To design and implement an automated backup solution using Azure Blob Storage and scheduling tools.

Azure Blob Storage is a cloud-based object storage service used to store large amounts of unstructured data such as files, backups, logs, and media. It provides high availability, scalability, and secure storage for backup and disaster recovery solutions.

Procedure

Step 1: Create Azure Storage Account

1. Login to Azure Portal: <https://portal.azure.com>
2. Click **Create a resource**
3. Search **Storage Account** → Click **Create**

Fill the following details:

- Subscription: Azure Student Subscription
- Resource Group: Create new (backup-rg)
- Storage Account Name: mystoragebackup123
- Region: South India
- Performance: Standard
- Redundancy: Locally Redundant Storage (LRS)
- 4. Click **Review + Create** → **Create**

Step 2: Create Blob Container

1. Open Storage Account
2. Go to **Data Storage** → **Containers**
3. Click + **Container**

Fill:

- Name: backups
- Public Access: Private
- 4. Click **Create**

Step 3: Azure CLI Login

```
az login
```

```
# Opens browser for authentication
```

```
az account show
```

```
# Verify account details
```

Step 4: Upload Files

```
az storage blob upload-batch \  
  --account-name mystoragebackup123 \  
  --destination backups \  
  --source C:\backup-data  
# Uploads all local files to Azure
```

Step 5: Create Backup Script

```
$source = "C:\backup-data"  
$container = "backups"  
  
az storage blob upload-batch `  
  --account-name mystoragebackup123 `  
  --destination $container `  
  --source $source  
# Automates backup process
```

Step 6: Add Timestamp Backup

```
$date = Get-Date -Format "yyyy-MM-dd"  
$destination = "backups/$date"  
  
az storage blob upload-batch `  
  --account-name mystoragebackup123 `  
  --destination $destination `  
  --source C:\backup-data  
# Stores backup by date
```

Step 7: Scheduling using Windows Task Scheduler (Detailed)

Steps:

1. Press **Windows + R**
2. Type:
taskschd.msc
3. Press Enter
4. Click **Create Basic Task**
5. Enter:
 - Name: AzureBackupTask
 - Description: Daily Azure Backup
6. Click **Next**

Set Trigger

- Choose: **Daily**
- Set time (e.g., 2:00 AM)
- Click Next

Set Action

- Choose: **Start a Program**

Configure Program

powershell.exe

Add Arguments:

-ExecutionPolicy Bypass -File "C:\scripts\backup.ps1"

Start In:

C:\scripts

Finish

- Click Next → Finish

Result

Successfully implemented automated backup using Azure Blob Storage with scheduling.